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Experiment 4: Provision a VPS instance from a cloud provider or set up a virtual machine on your local system using software like VirtualBox or VMware. 2. Install a Linux distribution such as Ubuntu, CentOS, or Debian as the operating system. Configure network settings, including IP address, subnet mask, gateway, and DNS.

AIM:

To provision a Virtual Private Server (VPS) using VirtualBox, install Ubuntu Server, configure network settings (IP Address, Subnet Mask, Gateway, DNS), and enable SSH remote access.

What is VPS?

A Virtual Private Server (VPS) is a software-based server created using virtualization technology. It operates independently with its own operating system, storage, memory, and network configuration while sharing the physical hardware with other virtual machines. Virtualization allows efficient utilization of resources and forms the foundation of cloud computing platforms.

Network configuration parameters include:

IP Address

A unique address assigned to a device on a network.

Subnet Mask

Determines the network and host portions of an IP address.

Gateway

Provides access to networks outside the local network.

DNS (Domain Name System)

Converts domain names into IP addresses.

SSH (Secure Shell)

A secure protocol used for remote administration of Linux systems.

PROCEDURE:

1. Open the Ubuntu terminal and verify the internet connection using the ping google.com command.

```
valid_lft forever preferred_lft forever
vboxuser@Ubuntu:~/Desktop$ ping google.com
PING google.com (142.251.221.206) 56(84) bytes of data:
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=1 ttl=64 time=35.8 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=2 ttl=64 time=36.1 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=3 ttl=64 time=64.2 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=4 ttl=64 time=51.4 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=5 ttl=64 time=60.5 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=6 ttl=64 time=239 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=7 ttl=64 time=256 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=8 ttl=64 time=93.1 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=9 ttl=64 time=48.6 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=10 ttl=64 time=34.1 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=11 ttl=64 time=55.8 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=12 ttl=64 time=51.9 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=13 ttl=64 time=33.4 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=14 ttl=64 time=149 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=15 ttl=64 time=246 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=16 ttl=64 time=64.7 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=17 ttl=64 time=81.8 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=18 ttl=64 time=48.2 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=19 ttl=64 time=44.5 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=20 ttl=64 time=264 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=21 ttl=64 time=122 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=22 ttl=64 time=50.8 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=23 ttl=64 time=48.9 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=24 ttl=64 time=52.6 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=25 ttl=64 time=34.0 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=26 ttl=64 time=56.1 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=27 ttl=64 time=60.1 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=28 ttl=64 time=148 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=29 ttl=64 time=45.3 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=30 ttl=64 time=64.6 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=31 ttl=64 time=277 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=32 ttl=64 time=42.7 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=33 ttl=64 time=36.0 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=34 ttl=64 time=241 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=35 ttl=64 time=301 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=36 ttl=64 time=37.5 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=37 ttl=64 time=58.2 ms
```

2. Check the network routing information using the ip route command.

```
valid_lft forever preferred_lft forever
vboxuser@Ubuntu:~/Desktop$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.20.20.0/24 dev br-ex proto kernel scope link src 10.20.20.1
```

3. Verify the DNS (Domain Name System) configuration of the system using the cat /etc/resolv.conf command.

```
vboxuser@Ubuntu:~/Desktop$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search .
```

4. Update the Ubuntu package list (if required) using:

```
sudo apt update
```

5. Install the OpenSSH Server package using:

```
sudo apt install openssh-server -y
```

6. Enable and start the SSH service using:

7. `sudo systemctl enable ssh`

```
sudo systemctl start ssh
```

8. Check whether the SSH service is running successfully using:

```
sudo systemctl status
```

```
Processing triggers for dmcc (0.36.2-9ubuntu1)
vboxuser@Ubuntu:~/Desktop$ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
```

9. Find the IP address of the Ubuntu system using:

```
Hostname -I
```

10. Establish an SSH connection to the Ubuntu system using:

```
ssh <username>@<IP_address>
```

```
Secure Shell (SSH)
vboxuser@Ubuntu:~/Desktop$ hostname -I
10.0.2.15 10.20.20.1 fd17:625c:f037:2:a00:27ff:fe5c:c094
vboxuser@Ubuntu:~/Desktop$ ssh vboxuser@10.0.2.15
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is: SHA256:DPutENGud479D6pbIQjtRgFAhR1uP74ZPyEInGKGpbA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.15' (ED25519) to the list of known hosts.
vboxuser@10.0.2.15's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

11. Verify the successful SSH login by checking the "Welcome to Ubuntu" message displayed on the terminal.

CONCLUSION:

The Ubuntu system was successfully configured and verified for network connectivity. The OpenSSH Server was installed and enabled, and the SSH service was confirmed to be running. The system's IP address was obtained, and a successful SSH login was performed. Thus, secure remote access to the Ubuntu system was successfully established and verified.